Bayesian Survival Methodology Design

with Informative Priors for Hydrogen Project Implementation Risk

Thesis Extension — Working Document

Sake Saakstra

MSc EOR Financial Track, Vrije Universiteit Amsterdam

Purpose. This document specifies a Bayesian survival analysis design that extends the frequentist Cox-and-Fine-Gray framework used in the paper. It targets three improvements: (i) principled treatment of small-event-count uncertainty through informative priors, (ii) coherent propagation of uncertainty across the carbon-conditional structure, and (iii) explicit incorporation of adjacent-literature evidence into the inferential framework. The document is intentionally structured around *prior choices that can be defended* rather than purely model implementation. Reviewers and supervisors should be able to evaluate each prior independently.

Version 0.1 — working draft for supervisor discussion — May 18, 2026

Contents

1 Motivation: why Bayesian, why now

The frequentist analysis in the working paper relies on 43 cancellation/on-hold events across 714 Blue_CCS and PEM projects. Several open issues arise from this sample structure:

1. **Wide confidence intervals on higher-order parameters.** The Fine-Gray on-hold HR (1.20, 95% CI [0.46, 3.11]) is informative about the absence of effect on delay but is too imprecise to support strong inferential claims. The carbon-conditional interaction (Blue $\times$ EUA $= -2.51$) has tight standard errors at the mean but extrapolates substantially at $z = \pm 1$ (HR $= 673$ vs 4.67), where the data is sparser.
2. **Information from adjacent literature is currently discarded.** The frequentist framework treats each estimate as if no prior knowledge exists. In practice, the literature on hydrogen project realization (Odenweller et al., 2022), CCS economics (Budinis et al., 2018; Mac Dowell et al., 2017), and transition-risk asset pricing (Bolton & Kacperczyk, 2021) all contain quantitative information that informs reasonable parameter ranges.
3. **Sensitivity to model specification is not transparent.** Frequentist robustness across eleven estimators demonstrates qualitative convergence, but does not give a single posterior that integrates all sources of uncertainty.

A Bayesian survival framework with informative priors addresses all three issues simultaneously. Importantly, it does so while remaining *methodologically conservative*: with weakly informative priors as the reference specification, the posterior collapses toward the frequentist MLE when data is sufficient and toward the prior only when data is genuinely uninformative.

Rationale

The case for Bayesian is not primarily about "better estimates" — it is about *transparent uncertainty propagation* and *principled use of prior knowledge*. In a setting with 43 events and explicit identification limitations, both properties have direct analytical value.

2 Model specification

The target model extends the carbon-conditional Cox formulation from the working paper:

$$\lambda_{ij}(t \mid \boldsymbol{X}_i, \mathrm{EUA}_t, \nu_j) = \lambda_0(t) \exp\big(\beta_1 \mathrm{Blue\_CCS}_i + \beta_2 \widetilde{\mathrm{EUA}}_t + \beta_3 (\mathrm{Blue\_CCS}_i \times \widetilde{\mathrm{EUA}}_t) + \boldsymbol{\gamma}^\top \boldsymbol{Z}_i + \nu_j\big),$$

where ν_j is a sponsor-level frailty (sponsor index j), $\boldsymbol{Z}_i$ contains controls (region fixed effects, sponsor-type categorical, log-scale), and all macro-financial covariates are standardized.

For the competing-risks extension, we apply the same structure separately to the subdistribution hazards $\bar{\lambda}_k(t \mid \cdot)$ for $k \in \{\text{cancelled}, \text{on-hold}\}$ in the Fine-Gray framework.

Technical note

Three implementation options for Bayesian fitting:

- `brms` (R wrapper around Stan) — supports Cox and Fine-Gray-like models via `family = cox`; good for accessible specification but limited customization of priors for the baseline hazard.
- `rstanarm` — pre-compiled Stan models, faster but less flexible.
- `Stan` direct — maximum flexibility, requires explicit hazard specification (piecewise or Weibull). Required for non-standard frailty structures.

Initial recommendation: start with `brms` for the Cox specification, move to direct Stan when implementing the Fine-Gray competing-risks model with shared frailty.

3 Prior specification, parameter by parameter

This section is the substantive content of the document. Each parameter receives: (a) a rationale for the chosen prior family, (b) a default *weakly informative* prior to be used as the reference specification, (c) an *informative* prior alternative, with literature backing, (d) explicit sensitivity analyses to be reported.

3.1 β_1 — main effect of Blue_CCS technology indicator

The frequentist Cox PH estimate is $\hat{\beta}_1 \approx 2.48$ (HR ≈ 11.9, 95% CI on HR $[4.67, 30.49]$). On the log-HR scale, the standard error is approximately 0.45.

Rationale

The literature suggests *some* expectation that blue hydrogen projects face higher realization risk than green, but quantitative anchoring is scarce.

- Odenweller et al. (2022) estimates electrolyser project realization rates around 30–40%, with no parallel estimate for blue hydrogen with CCS. Inference: blue hydrogen has *at least* comparable risk, plausibly higher.
- Budinis et al. (2018) and Mac Dowell et al. (2017) document well-known barriers to CCS deployment (cost overruns, CCS infrastructure unavailability, public acceptance). These translate into elevated project cancellation risk relative to renewable-electricity-based pathways.
- Bolton & Kacperczyk (2021) and Caldecott (2018) frame fossil-derivative projects as carrying "transition risk" that is partially priced into firm valuations, suggesting some prior on a non-zero blue-disadvantage.

None of this literature gives a precise quantitative anchor. We therefore use moderately informative priors centered away from zero but with substantial uncertainty.

Prior	Specification	Implies HR (median, 95% range)
Vague (reference 1)	$\beta_1 \sim \mathcal{N}(0, 5)$	1, [0.0001, 22000]
Weakly informative (default)	$\beta_1 \sim \mathcal{N}(0, 2)$	1, [0.02, 50]
Skeptical	$\beta_1 \sim \mathcal{N}(0, 1)$	1, [0.14, 7.4]
Informative	$\beta_1 \sim \mathcal{N}(1.5, 0.7)$	4.5, [1.1, 18.5]

The informative prior is centered at HR ≈ 4.5, somewhat below the frequentist point estimate of HR ≈ 12. This is deliberate: it represents the strength of the literature-based expectation prior to seeing the IEA data. Posterior updating will move toward higher HR if the data supports it.

3.2 β_2 — main effect of standardized EUA carbon price

The frequentist estimate is approximately $\hat{\beta}_2 \approx -0.3$, with mild precision. On its own, this coefficient captures the marginal effect of EUA on the PEM baseline (since the interaction with Blue_CCS is separate).

Rationale

Theoretical priors:

- PEM economics are dominated by electricity costs, not EUA exposure directly. Therefore the EUA main effect on PEM is plausibly small.
- Indirect channels exist: higher EUA → higher industrial gas prices → higher demand for hydrogen substitutes (helps PEM) → lower cancellation risk.
- Plausible prior expectation: slightly negative effect (higher EUA → slightly lower cancellation hazard), small magnitude.

Prior	Specification	Implies HR per 1 SD EUA increase
Vague	$\beta_2 \sim \mathcal{N}(0, 5)$	1, [0.0001, 22000]
Weakly informative (default)	$\beta_2 \sim \mathcal{N}(0, 1)$	1, [0.14, 7.4]
Informative	$\beta_2 \sim \mathcal{N}(-0.2, 0.5)$	0.82, [0.31, 2.18]

3.3 β_3 — the Blue × EUA interaction (key parameter)

This is the parameter that drives the carbon-conditional hazard finding. The frequentist estimate is $\hat{\beta}_3 \approx -2.51$, $p < 0.0001$.

Rationale

This is methodologically the most consequential prior. Two considerations:

- **Skeptical prior is the conservative choice.** A prior centered at zero ("no interaction effect") requires the data to overcome the prior. With the strong frequentist signal ($p < 0.0001$), this prior would lead to posterior estimates somewhat attenuated toward zero, which is methodologically defensible.
- **Informative prior risks circular reasoning.** The informative prior should be set based on *ex-ante* reasoning, not on the data we have already analyzed. The ex-ante literature does suggest carbon-conditional effects on transition technology (Bolton & Kacperczyk; insurance interpretations in Caldecott), but does not anchor magnitude.

We recommend the *skeptical prior* as the default reference, and a *vague prior* as one robustness check. We do *not* use a strongly informative prior on this parameter because of the circularity concern.

Prior	Specification	Implies interaction effect range
Vague (robustness)	$\beta_3 \sim \mathcal{N}(0, 5)$	Wide; lets data dominate
Weakly informative (default 1)	$\beta_3 \sim \mathcal{N}(0, 2)$	95% in $[-4, 4]$
Skeptical (default 2)	$\beta_3 \sim \mathcal{N}(0, 1)$	95% in $[-2, 2]$

Open question for supervisor

Should we report *both* the weakly informative and the skeptical prior as primary specifications, or pick one? My current view is that reporting both transparently (as primary + sensitivity) is the most honest — it shows readers how much the conclusion depends on prior choice. Bos's view?

3.4 γ — control variable effects (region, sponsor type, log-scale)

These are nuisance parameters in our setting; we just need them not to soak up parameter mass we want to allocate elsewhere.

- Region fixed effects: $\gamma_r \sim \mathcal{N}(0, 1.5)$ each. Implies effects $\leq$ HR ≈ 20 vs base region with 95% mass.
- Sponsor-type categorical (e.g., supermajor, mid-cap, public utility, JV): $\gamma_s \sim \mathcal{N}(0, 1.5)$ each.
- Log-scale (continuous): $\gamma_{\text{scale}} \sim \mathcal{N}(0, 1)$.

3.5 Sponsor-level frailty variance θ

In the frequentist specification, the shared frailty Cox estimated $\hat{\theta} \approx 0$, suggesting no detectable sponsor-level heterogeneity beyond observable covariates. This is plausible given that sponsor-type is included as a fixed effect, and most sponsors have a small number of projects.

Rationale

With limited within-sponsor variation, the frailty variance is weakly identified. The prior choice matters less for the headline result but matters for whether the model converges sensibly.

- Default: $\theta \sim \mathcal{N}^+(0, 1)$ — weakly informative, allows up to $\theta \approx 2.5$ within 95% mass.
- Robustness 1: $\theta \sim \text{HalfCauchy}(0, 1)$ — heavier tails, more conservative.
- Robustness 2: $\theta \sim \mathcal{N}^+(0, 0.3)$ — more concentrated on near-zero values, encoding the frequentist finding of negligible frailty.

3.6 Baseline hazard $\lambda_0(t)$

In a fully Bayesian Cox model we need to specify the baseline hazard. Several options:

1. **Weibull parametric**: $\lambda_0(t) = \alpha\lambda t^{\alpha-1}$. Smooth, two parameters. Implies hazard is monotonic in t (decreasing if $\alpha < 1$, increasing if $\alpha > 1$). Project cancellation hazard is likely *not* monotonic — it tends to peak in early years.
2. **Piecewise constant**: split observation period into K intervals, with $\lambda_{0,k}$ for each interval. Flexible, popular default for Bayesian survival.
3. **Spline-based**: smooth nonparametric baseline. More flexible but harder to interpret.

Recommendation: piecewise constant with $K = 5$ intervals (yearly bins from project announcement to data cutoff), with each $\lambda_{0,k} \sim \text{Gamma}(0.5, 0.5)$. This is the `brms` default for piecewise hazards.

Open question for supervisor

The choice of K (number of intervals) is a hyperparameter. Larger K gives more flexibility but worse identifiability. With 43 events, $K = 5$ feels right — about 8 events per interval. Bos: thoughts on whether this should be tuned more carefully, perhaps via cross-validation on out-of-sample log-likelihood?

4 Competing-risks extension

For the Fine-Gray competing-risks model, the structure replicates but with two parameter vectors:

- $\beta^{(\text{cancel})}$ for the cancellation subdistribution hazard

- $\boldsymbol{\beta}^{(\text{on-hold})}$ for the on-hold subdistribution hazard

A natural prior structure couples the two parameter vectors via a shared baseline expectation, while allowing them to differ where the data supports it. This can be implemented as:

$$\beta_1^{(\text{cancel})} \sim \mathcal{N}(\mu, \tau_1^2), \quad \beta_1^{(\text{on-hold})} \sim \mathcal{N}(\mu, \tau_1^2),$$

with μ, τ_1 themselves hierarchical hyperpriors. Alternatively, we use independent priors as in the cancellation-only specification, and let the data show whether the two are different.

Rationale

The frequentist Fine-Gray results show $\text{HR}^{(\text{cancel})} = 13.19$, $\text{HR}^{(\text{on-hold})} = 1.20$. The difference is the substantive finding (terminal vs delay). In a Bayesian setting we want to test whether this difference is robust to prior choice. Coupling the priors would tend to pull them toward each other; independent priors let the data decide.
Initial recommendation: **independent priors** for primary specification; **coupled hierarchical priors** as one sensitivity analysis.

5 Sensitivity analysis plan

For each parameter β_j in the model, we plan to report posterior summaries under three prior specifications:

1. **Reference (weakly informative)** — the default specification listed above.
2. **Skeptical** — tighter priors centered at zero.
3. **Informative** — centered at literature-based expectations (where defensible).

For the headline interaction β_3 specifically, we will also report under the **vague** prior to show that the data overwhelms even very loose priors. The full sensitivity table will be a $3 \times K$ grid (specification by parameter), reported as the central result.

Rationale

This is a stronger transparency move than typical frequentist sensitivity tables, which are usually limited to "add/drop covariate." Bayesian sensitivity over prior choice is the more methodologically defensible analog of frequentist robustness over model specification.

6 Prior predictive checks

Before fitting models to data, we will perform *prior predictive simulation*: generate hypothetical project-cancellation patterns from each prior specification (with no data), and check whether they are reasonable.

Specifically: for each prior set, draw $S = 1000$ values of $(\beta_1, \beta_2, \beta_3, \gamma, \theta, \lambda_0)$, and simulate hypothetical IEA-like project histories. Check that:

- The implied range of cancellation rates is plausible (say, 1%–50% over 5 years).
- The implied range of HR values is plausible (say, 0.1–100).
- The carbon-conditional HR at $z = \pm 1$ is finite and plausible.

If a prior fails these checks (e.g., implies most projects cancel within months, or implies HR $> 10^6$), it is rejected on prior-predictive-validity grounds.

7 Posterior diagnostics

For each fitted model we will report:

- Trace plots and effective sample size for all parameters
- $\hat{R}$ (Gelman-Rubin) convergence statistic; target $\hat{R} < 1.01$
- Pairs plots for parameter correlations (especially β_1, β_3)
- Posterior predictive checks: simulate from posterior, compare to observed cancellation patterns
- Leave-one-out cross-validation (LOO-CV) for model comparison

8 Implementation roadmap

1. **Week 1–2.** Replicate frequentist Cox PH and Fine-Gray in `brms` with weakly informative priors. Verify that posterior modes align with frequentist MLEs (sanity check).
2. **Week 3–4.** Add informative-prior specifications. Implement sensitivity grid. Run prior predictive checks.
3. **Week 5–6.** Implement competing-risks Bayesian Fine-Gray (likely via direct Stan code, as `brms` support is limited).
4. **Week 7–8.** Run on extended IEA September 2025 dataset (if available). Write methodology chapter for thesis.
5. **Week 9–10.** Sensitivity analyses, posterior diagnostics, write-up.

9 Open questions for supervisor discussion

Open question for supervisor

Q1. For the headline interaction β_3, should we present skeptical and weakly informative priors as co-primary, or choose one?

Open question for supervisor

Q2. Baseline hazard: piecewise constant with $K = 5$ vs Weibull vs spline? The right choice may depend on whether the thesis emphasizes interpretability or flexibility.

Open question for supervisor

Q3. Should the Fine-Gray competing-risks Bayesian model use independent priors per event type, or coupled hierarchical priors? Substantive tradeoff vs methodological elegance.

Open question for supervisor

Q4. Is the planned sensitivity grid (3 priors $\times$ K parameters) appropriately scoped, or should we add specific extra priors (e.g., priors elicited from industry experts at Gasunie or S&P)?

Open question for supervisor

Q5. Implementation: `brms` vs direct Stan. `brms` is faster to set up but limited for competing-risks. Direct Stan gives full flexibility but more development time. Recommendation?

Open question for supervisor

Q6. Should we explore state-space versions where $\beta_3(t)$ is itself time-varying (Koopman-style), or is that scope creep for the MSc thesis (and better reserved for PhD)?

10 References (working list, to be expanded)

- Bolton, P., & Kacperczyk, M. (2021). Do investors care about carbon risk? *Journal of Financial Economics*, 142(2), 517–549.
- Budinis, S., et al. (2018). An assessment of CCS costs, barriers and potential. *Energy Strategy Reviews*, 22, 61–81.
- Caldecott, B. (2018). Stranded assets: the role of climate disclosure. *Journal of Sustainable Finance & Investment*.
- Fine, J. P., & Gray, R. J. (1999). A proportional hazards model for the subdistribution of a competing risk. *JASA*, 94(446), 496–509.
- Gelman, A., et al. (2017). The prior can often only be understood in the context of the likelihood. *Entropy*, 19(10), 555.
- Mac Dowell, N., et al. (2017). The role of CO_2 capture and utilization in mitigating climate change. *Nature Climate Change*, 7, 243–249.
- Odenweller, A., et al. (2022). Probabilistic feasibility space of scaling up green hydrogen supply. *Nature Energy*, 7(9), 854–865.
- Vehtari, A., et al. (2021). Rank-normalization, folding, and localization: an improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2), 667–718.
- Bauwens, L., Bos, C., & Van Dijk, H. (1999). Adaptive Polar Sampling with an Application to a Bayes Measure of Value-at-Risk. *Tinbergen Institute Discussion Paper*.
- Creal, D., Koopman, S. J., & Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.